

Information Visualization Project Proposal

February 27, 2025

Alex Caramanico, Wooje Chang, Benjamin Knight, Bahereh Vojdani
CMSC 734 - Leo Zhicheng Liu

1. *Problem*

Precise prediction of carbon dioxide (CO₂) emissions from buildings is essential for improving sustainable energy practices and mitigating environmental damage. A **prediction model** for calculating CO₂ emissions in 140 buildings on the University of Maryland, College Park (UMD) campus **requires accurate and meaningful insights**. One of the most intuitive and natural ways to develop these insights is the use of a visual aide. Such an aide can provide quick information on the relationships between variables, recent trends, and reveal clusters of similar buildings. All of which can be used as either direct inputs or as useful guidance for the development of the prediction model. Such a model would employ a synthesis of actual building data and simulated solar radiation data - elements the visualization seeks to simplify, contextualize, and effectively communicate. The main goal of this project is to provide useful perspectives for energy management and sustainability efforts, in order to drive a more environmentally conscious campus by providing a tool for professionals to verify and communicate contributions made towards sustainability.

2. *Users*

The primary users are architects and engineers, who can use the tool to design renovations or entirely new buildings which seek to improve sustainability. Their expertise lies in design and construction, and a visualization aid such as ours can be helpful in blueprinting new projects or renovating old ones. The tool can reveal specific relationships between variables, which buildings are currently performing well, and which are not. Engineers can use this information to guide future decisions. The necessity is clear - carbon dioxide emissions are a complicated but pertinent issue in the move toward sustainability which will improve quality of life for everyone. Thus, any device which may streamline (accelerate, and/or simplify) this process is of great use. A visualization specifically helps because of its speed, intuitiveness, and flexibility. Other methods such as lengthy literature reviews can be time-consuming and relatively inflexible as new information becomes available. However, a visualization tool can be made to accommodate for new data over time, and is relatively simple and quick to fine-tune to achieve optimal utility. A visualization tool also simplifies complex relationships between variables in a way that can be understood in a matter of seconds, and potentially opens the door for automated procedures or some sort of “flagging”

system when certain parameters stray from an acceptable range. Since this group has knowledge on construction techniques, they can effectively leverage the information conveyed from the visualization to account for sustainability efforts in an efficient way.

An obvious secondary group that could potentially gain equal benefit from our project are policymakers and stakeholders. The utility of a carbon dioxide visualization tool is clear to them as well - knowledge of carbon dioxide trends and its relationship to other variables can influence decision making for *legislators* and *building owners* who may be called to action upon seeing recent trends. This could involve renovations and/or increased regulations on the activities that take place in said building. The main expertise of the secondary group lies within the social and economic factors involved with construction legislation and building regulations/policies. They would be expected to understand *how* to use the information provided by the tool, rather than the inner workings of the tool itself. These people may possess less precise knowledge on the scientific intricacies of carbon dioxide emissions, but require a working understanding so that they can make informed decisions to support their constituents (in the case of policymakers) or the people using their buildings (in the case of stakeholders). Specifically, the visualization can assist in budgeting for policymakers and in regulating for stakeholders. Obviously, the latter wants people to use their building so they can maintain a profit. The consumers' desire is largely dependent upon the quality of said building, which is a function of its sustainability.

Our team does not necessarily have direct access to either user group - though consulting with architects and engineers is tangentially possible through communication with students in said programs. These people could easily be reached through channels provided by the University of Maryland. Additionally, if the goal was to market our visualization, it would be entirely possible to design an experiment where the user group compares our methodology to their current standards to assess its efficacy. We assert that both groups can equally benefit from our proposed visualization aide, thus both are included in this discussion.

It should also be noted that the project is greatly generalizable to actual cities because the UMD campus is considered a small city. Thus, the insights and solutions provided by our visualizations can be applied in a wide range of contexts. We would argue that the tool has potential for very high external utility.

3. *Goal and tasks*

Goal	Improve University of Maryland sustainability by developing a CO2 data visualization tool				
Tasks	Establish variable relationships - potential prediction model input	Assess evolution of CO2 emissions over time, relate to static data	Contextualize benchmark EUI with current EUI	Identify sub-variables with greatest	Identify building clusters to study causal

							contribution to CO2 emissions	relationships
Sub-Tasks	Identify significant correlations to CO2: building area, construction date, building type, solar radiation, energy consumption	Aggregation of variables - buildings into size groups, average CO2 for building type. Account for aggregation with plotting methods (std error bars), visualize ANOVA	Times-series plots revealing trends, useful to predict the future	Establish whether buildings with specific trends (group them together) have static variables reflecting that	Split data into groups - buildings that have met benchmark and buildings that have not	Determine which variables may be responsible. Trends from time series plots should corroborate a “good” EUI building	i.e. which of natural gas, electricity, steam affects CO2 the most for a given building (could do visualizations of principal component analysis)	Determine which factors link buildings together the most, contextualize with findings from other tasks (a la factor analysis)

4. Dataset info

a. Describe schema

i. “Static” Data

The data is in a simple tabular format - each column representing a variable, each row representing a building on the UMD campus. The initial data is split into separate tables, one containing solar radiation information and the other containing more detailed information on energy consumption (and over thirty additional buildings). These will be joined together (by matching buildings) so that the “master” data set will contain much more information to be used for the tasks described above. Some of the included variables are listed below:

- Nominal
 - Building Name (Ritchie Coliseum, Atlantic Building, etc)
 - Building Type (Lab, Office, Residential, School, etc)
- Quantitative
 - Total CO2 Emissions (kkg)
 - Total electricity (kWh)
 - Area (sq feet)
 - Solar radiation (for each of the four seasons - simulated)
 - Energy usage intensity (EUI)
 - EPA benchmark
- Ordinal

- Year built

ii. Time Series Data

This data is also in a simple tabular format, the rows representing unique buildings and the columns indicating the total CO2 consumption (in kBtu) for each building for each month from February 2020 to January 2024.

Each variable is of course quantitative aside from the building names, which are nominal. There is also a Building ID variable, which may be useful in some ways for programming but is redundant of building name.

b. Sample data items

For convenience, the following statistics and charts were generated using R and strictly serve as an initial demonstration of the data. They may not necessarily be a significant part of the final product.

- *Static Data*

Summary statistics

Total (kkg CO2e)	Total (kwh)_Chiller Electric	Total (kwh)_Electricity	Total (Therms)_Natural Gas
Min. : 0	Min. : 0	Min. : 0	Min. : 0
1st Qu.: 49879	1st Qu.: 0	1st Qu.: 462850	1st Qu.: 0
Median : 219137	Median : 0	Median : 1240707	Median : 0
Mean : 1158870	Mean : 1596826	Mean : 5031681	Mean : 15833
3rd Qu.: 1219492	3rd Qu.: 685624	3rd Qu.: 5171738	3rd Qu.: 6924
Max. : 14356541	Max. : 33109092	Max. : 52270604	Max. : 421453

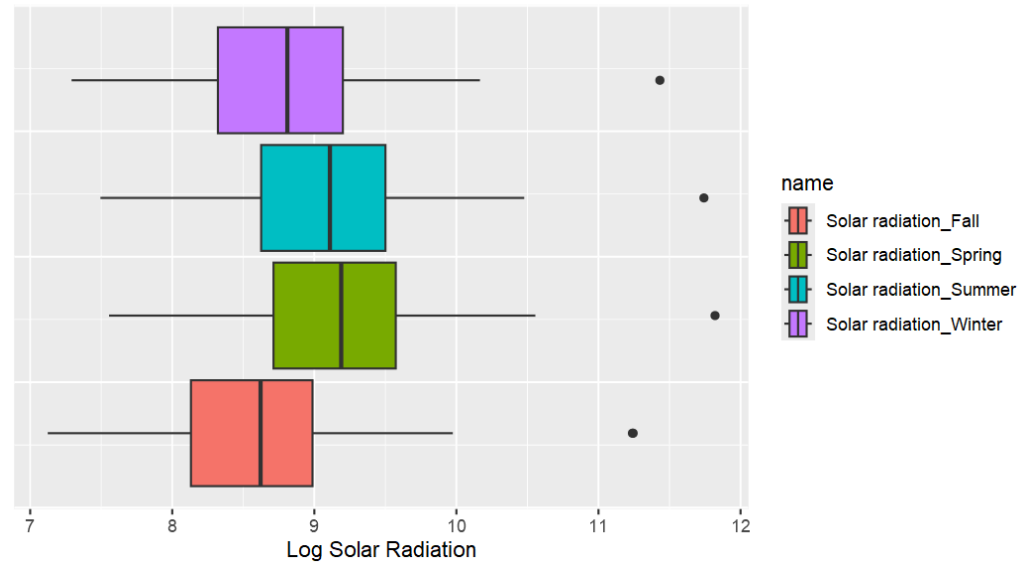
Total (lbs)_Steam	Total (kBtu)_Total Energy	Area(sq ft)	Year Built	UMD Type
Min. : 0	Min. : 182670	Min. : 1131	Min. : 1909	Classroom : 2
1st Qu.: 0	1st Qu.: 3958045	1st Qu.: 11700	1st Qu.: 1954	Lab : 36
Median : 1396216	Median : 10585122	Median : 33330	Median : 1963	Office : 40
Mean : 12339861	Mean : 38729379	Mean : 78468	Mean : 1967	Other : 31
3rd Qu.: 13646111	3rd Qu.: 38878437	3rd Qu.: 113328	3rd Qu.: 1990	Residential : 61
Max. : 164820665	Max. : 468275731	Max. : 524031	Max. : 2018	School : 2

Year Renovated	Annual EUI	EPA Benchmark EUI	Solar radiation_Winter	Solar radiation_Spring
Min. : 1952	Min. : 0.00	Min. : 23.00	Min. : 1468	Min. : 1911
1st Qu.: 1985	1st Qu.: 49.50	1st Qu.: 53.00	1st Qu.: 4111	1st Qu.: 6079
Median : 1992	Median : 71.00	Median : 61.00	Median : 6702	Median : 9790
Mean : 1993	Mean : 93.16	Mean : 65.03	Mean : 8487	Mean : 12248
3rd Qu.: 2003	3rd Qu.: 114.00	3rd Qu.: 70.00	3rd Qu.: 9906	3rd Qu.: 14373
Max. : 2017	Max. : 427.00	Max. : 216.00	Max. : 92236	Max. : 135991
NA's : 91	NA's : 1		NA's : 34	NA's : 34

Solar radiation_Summer	Solar radiation_Fall
Min. : 1798	Min. : 1241
1st Qu.: 5573	1st Qu.: 3400
Median : 9042	Median : 5551
Mean : 11390	Mean : 7454
3rd Qu.: 13365	3rd Qu.: 8000
Max. : 125767	Max. : 76401
NA's : 35	NA's : 34

Solar Radiation

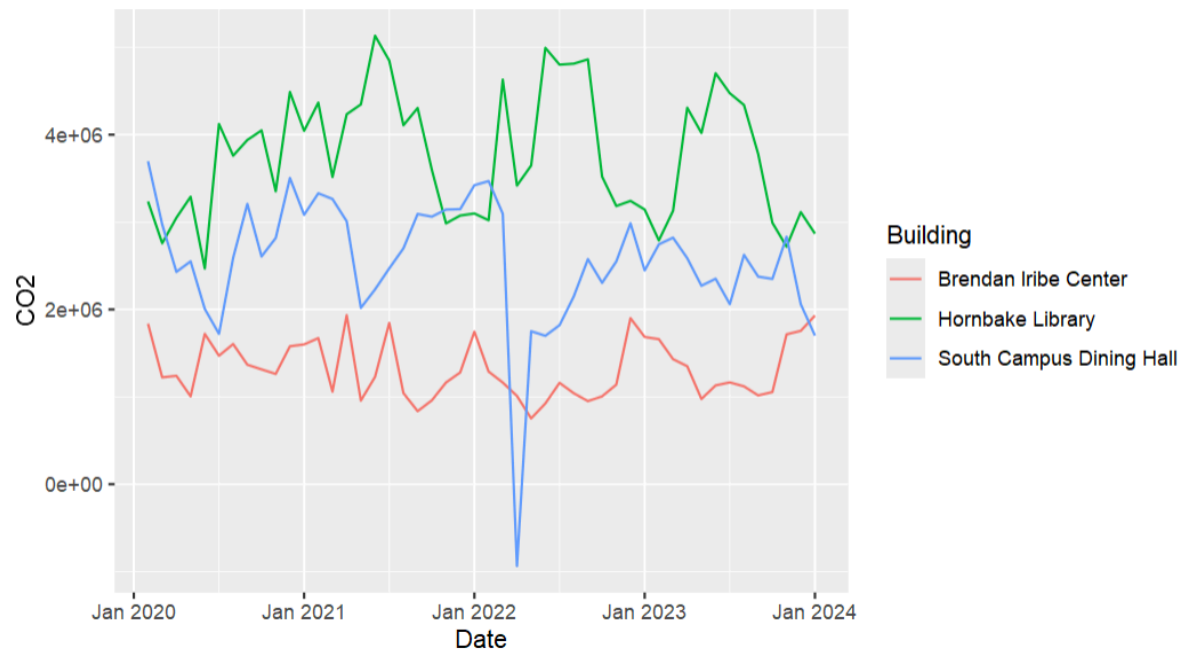
NOTE: Extreme outliers



Time Series Data

Carbon Dioxide (kBtu) Over Time

University of Maryland Campus



c. Size of data

i. Merged Data

The solar radiation data is 139 rows and 12 columns, while the more detailed data is 172 rows and 14 columns. After matching these data sets

together by building and removing the redundant variables, the final “static” data set is 172 rows and 18 columns. The buildings missing from the solar radiation data (that are present in the more detailed data) will have *NA* values for the solar radiation variables.

ii. Time Series Data

The time series data set is 176 rows of 60 variables. Two of the variables are building identifiers, while the remaining are the total CO₂ emissions of each building at monthly time points. It should be noted that Leonardtown 240-243 have *NA* values beginning in August 2023 as their demolition began in May of that year.

d. Source

We found the [TerpFootprint](#) website, a UMD dashboard application that enables access to campus utility usage (at the building level). It includes information on electricity, gas, steam, etc. A zip file containing all the data will be submitted along with this document, while a link to the solar radiation data is [here](#).

5. *Related work and Justification*

Rationale

University campuses play a crucial role in the realm of sustainability. Sustainable campuses function as living labs where sustainable practices and theories are both taught and actively applied and observed. Ali and Anufriev (Ali & Anufriev, 2020) assert that the notion of a green university is rooted in environmental sustainability, which includes the judicious utilization of natural resources, proficient environmental management, pollution mitigation, and economic factors related to profit and cost within the overarching framework of the environment, economics, and society. Velazquez et al. (Velazquez et al., 2006) define a sustainable university as "a higher educational institution, either in entirety or in part, that addresses, engages, and advocates for the reduction of adverse environmental, economic, social, and health impacts resulting from resource utilization, in order to fulfill its roles in teaching, research, outreach, partnership, and stewardship, thereby assisting society in transitioning to sustainable lifestyles." Alshuwaikhat and Abubakar (Alshuwaikhat & Abubakar, 2008) assert that "A sustainable university campus must embody a healthy environment, foster a thriving economy through energy and resource conservation, minimize waste, implement effective environmental management, and advocate for equity and social justice in its operations, while disseminating these values at community, national, and global levels."

Background Information and Connections

Previous research studies have examined the explanatory elements influencing building energy consumption and sought to establish a link between energy consumption and both environmental and non-environmental factors. A study revealed a considerable

positive link between building energy use and building age, while outdoor climate exhibited a minor impact on energy use (Sekki et al., 2015). Conversely, another study (Jafary et al., 2016) suggested a strong correlation between energy use and outside temperatures. A variety of energy management strategies, including the activation and deactivation of heating systems and the implementation of an occupancy schedule for these systems, were utilized in a university building, resulting in a claimed 40% reduction in energy loads (Mata et al., 2009). A separate study (Chung & Rhee, 2014) identified a possible energy savings range of 6%–30% with the implementation of various energy-saving strategies, including adjustments to indoor temperatures, occupant behaviors, and building envelope characteristics. One study (Chow et al., 2003) demonstrated using modeling methodologies that implementing energy-efficient technologies and equipment might yield a 20% to 40% potential for energy savings.

Existing CO2 Visualization Methods

There are a number of pre-existing methods that have been used to analyze carbon dioxide data. The first uses animations to show the [progression of carbon dioxide over time](#). It is an effective use of time series data, utilizing a form of heat map in addition to line charts. This not only shows trends over time, but areas with the highest concentration of CO2 emissions. Perhaps we could take inspiration from this and create a method to animate the CO2 emissions of each building over time - on a heat map as well as a line chart. The next is an effective means of demonstrating the [magnitude of CO2 emissions](#). This visualization takes the form of a simple game, where the player drives around and observes enormous blocks representing various amounts of carbon dioxide. It greatly emphasizes the effectiveness of interactivity in data visualization. We could also take inspiration here - if we deem it important to adequately represent just how much CO2 has been emitted by the buildings. It could be especially useful as it relates to comparing the buildings. An interesting idea would be to overlay the size of the building with the size of its carbon footprint, which would reveal very inefficient small buildings or particularly efficient large buildings. Another interesting visualization uses a [radial chart while encoding CO2 emissions per capita and population](#). This is a very aesthetically pleasing visualization, which in itself might invite readers to dig deeper. This is easily applicable to our task, as we could replace the population parameters with building areas. We also have a number of other variables to work with which could reveal additional interesting patterns.

Further literature review reveals that many of the existing visualizations do not involve a great deal of *variable relationships*, which is a gap our project can fill. Most seem to be primarily concerned with gross numbers with relatively few derived values or comparisons to other variables which may contribute to the problem. Our proposal can greatly enrich the understanding of and the response to carbon dioxide emissions.

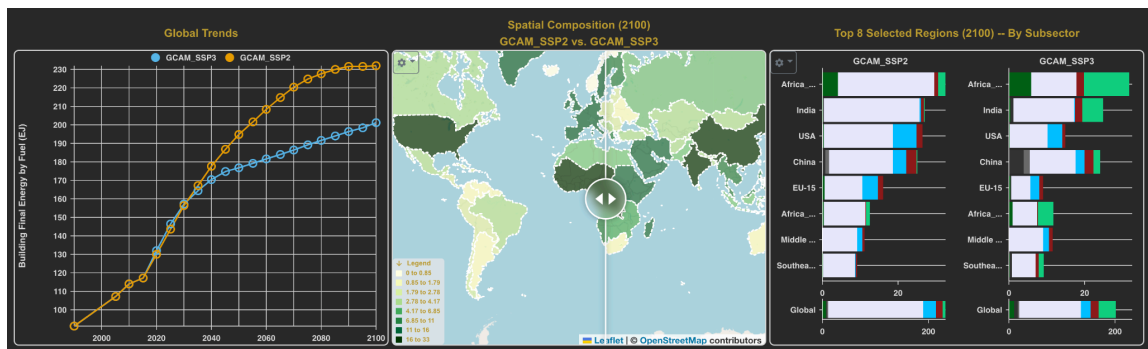
In addition to the visualizations discussed previously, there are a plethora of existing softwares to assist in such visualizations, such as Tableau, Data Formulator, the ggplot2 library in R, and several others. We attempt to provide a novel approach to the visualization of carbon dioxide and climate data by building upon what already exists. Additionally, as stated previously, the primary goal is for the visualization to provide useful insights for CO2 prediction models. This additional context will guide our decision making.

Our Own Work

One of our group members is currently working at PNNL where he is experienced with working with Integrated Assessment Models (IAMs), specifically the Global Change Analysis Model (GCAM). He has developed an interactive dashboard allowing the visualization of complex datasets from these models across energy, water, land use, and emissions. GCAM runs in both historical and predictive modes in five year intervals from 1990 to 2100. GCAM has various assumptions about policies, future technologies, and economic development and gives a wide range of outputs across climate effectiveness including emissions outputs from buildings at a 0.5x0.5 spatial resolution.

Our group member is working on Foresight, a data visualization platform designed to simply and effectively visualize complex datasets across various spatial scales, time, and aggregation levels from the GCAM model. A dynamic DynamoDB backend allows storage and quick querying of very large datasets while a flexible React.js frontend allows the utilization of a wide variety of visualization tools built off of d3. Existing experience in development of dashboards with related data will guide our choice in visualizations and database structure.

The Foresight GCAM Data visualization platform displaying CO2 Emissions by building type at a regional spatial scale:



GCAM Data can be used to validate assumptions of emissions from buildings as well as giving our team a valuable community of scientists to seek input and efficacy verification.

6. *Team*

Name	Role 1	Role 2	Role 3
Alex Caramanico	Initial dataset exploration/visuals	Web page and report write-ups, light programming	Statistical analysis of data and results, suggesting updates
Wooje Chang	Data proofing and formatting	Statistical analysis of data and results	Visualization creation of the analysis results
Benjamin Knight	Data extraction / formatting	Database creation / Data aggregation. Programming.	Visualization creation / development. Verification / Contextualization.
Bahereh Vohjdani	Background research and contextualization	Efficacy verification (assessing effectiveness)	Record/script video

References

- Ali, E. B., & Anufriev, V. P. (2020). *UI greenmetric and campus sustainability: A review of the role of African universities*.
- Alshuwaikhat, H. M., & Abubakar, I. (2008). An integrated approach to achieving campus sustainability: assessment of the current campus environmental management practices. *Journal of Cleaner Production*, 16(16), 1777–1785.
- Brewer, J., (March 10, 2021). Seeing CO2 brings data visualisation into a game world to make infographics playable
- Chow, S., Ganji, A. R., Hackett, B., Parkin, P., & Fethers, A. (2003). Energy assessment of selected schools in Anchorage School District. *26th World Energy Engineering Congress, Atlanta, GA*, 12–14.
- Chung, M. H., & Rhee, E. K. (2014). Potential opportunities for energy conservation in existing buildings on university campus: A field survey in Korea. *Energy and Buildings*, 78, 176–182.
- Jafari, M., Wright, M., Shephard, L., Gomez, J., & Nair, R. U. (2016). Understanding campus energy consumption--people, buildings and technology. *2016 IEEE Green Technologies Conference (GreenTech)*, 68–72.
- Kostis, HN., Ott, L., & Weir, B. (June 20, 2023). Global Atmospheric Carbon Dioxide (CO2) *National Aeronautics and Space Administration (NASA)*.
<https://svs.gsfc.nasa.gov/5115#:~:text=The%20visualization%20featured%20on%20this,can%20be%20seen%20building%20up>.
- Sekki, T., Airaksinen, M., & Saari, A. (2015). Measured energy consumption of educational buildings in a Finnish city. *Energy and Buildings*, 87, 105–115.
- Smith, M. (2021) Carbon Emissions per-capita by Country
<https://www.visualcapitalist.com/visualizing-global-per-capita-co2-emissions/>
- Velazquez, L., Munguia, N., Platt, A., & Taddei, J. (2006). Sustainable university: what can be the matter? *Journal of Cleaner Production*, 14(9–11), 810–819.